Emanuele La Falce

Aerospace Engineer



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Languages

Italian	Mother tongue
English	C2
Spanish	A2

Technical skills

STK (Systems Tool Kit) | MATLAB | C Programming | Python (basic) | MSC PATRAN/NASTRAN | LaTeX | SolidWorks Visualize

Honours and awards

ActInSpace 2022 – I3P Special Award
SpaceTug concept selected for a CNES-run startup pre-incubation path.

AESA Hackathon 2022 – Winner
Innovative drone use-case design; team decision-making.

CWMUN 2018 – Honourable mention (Diplomatici)

Volunteering

Buffoni di Corte – Volunteer (2024–Current): workshops with people with cognitive disabilities

Education

- 03/2023 - 07/2026 | POLITECNICO DI TORINO, Torino, Italy

  Master's Degree in Aerospace and Astronautical Engineering

- 01/07/2025 - 23/12/2025 | VIRGINIA TECH – United States, Virginia

  Visiting scholar - researcher

  Research on the **Three Body Problem dynamics**, **space flight** dynamics and **trajectory design** for my Master's thesis. Improving the trajectory for a mission in the Earth-Moon system (Space tug project) combining a low thrust phase with the use of manifold coasting up to the Lunar Gateway orbit. Trade-offs have been used considering both time-optimal and fuel-optimal scenarios.

- 09/2019 - 03/2023 | POLITECNICO DI TORINO

  Bachelor's Degree in Aerospace Engineering

  Thesis: "Theory and history of the Gravity Assist"

Professional experience

- 10/2026 - CURRENT | POLITECNICO DI TORINO

  Research Engineer

  Working on the **phasing problem**, trajectory and **mission design** for a Lunar mission's trajectory (Space Tug project)

- 09/2022 - 01/2023 | POLITECNICO DI TORINO

  Laboratory assistant

  Fluid mechanics laboratory assistant

Student teams experience

- 08/2023 - 09/2024 | ICARUS POLITO - (POLITECNICO DI TORINO)

  Project Manager

  Built and managed a team designing a hybrid rocket engine from the ground up (9000 m apogee); led mission analysis and configuration decisions during the design phase.

- 10/2022 - 09/2024 | Icarus PoliTo (POLITECNICO DI TORINO)

  Team Member

  Designed, developed and tested the propulsion and recovery systems for a solid-propellant rocket model (aim 3000 m).

Courses

- 2024 | Politecnico di Torino - Orbital Mechanics
- 2023 | Mission Project (Politecnico di Torino)

  Role: **Mission analyst**

  Worked as mission analyst for a group project and designed a satellite constellation to detect wildfires. Worked on *collision avoidance* with **Spice/Mice**, *orbit propagation and Space Situational Awareness* with **STK**
- 2022 | MIT OpenCourseWare – Introduction to Astronomy
- 2022 | MIT OpenCourseWare – Intro to Computer Science & Programming in Python